ENGS 104 Optimization Methods for Engineering Applications
Fall 2026
Final Project

Overview

The culminating experience in this course is a project in which you apply your knowledge of optimization to problems outside the scope of the course lectures. Your task is to identify a decision-making problem, encode it as an optimization problem, design or implement a solver, and analyze properties of the program and solution. Students are encouraged to select their own topic (which may relate to their individual research goals or engineering interests) and iterate on their project idea by meeting with the course instructor. Successful projects will include an analysis component, a computational component, and a discussion component; however, you may choose to put more emphasis on a specific component. Possible project structures include

1. *Literature based project* - Reproduce the optimization formulation and solution from a published paper, then test a modification to the model or method and analyze the effect.
2. *Problem formulation project* - Encode a real-world problem as an optimization problem, solve it computationally, and discuss the modeling choices and approximations made.
3. *Solver design project* - Implement a solver, test it across problem instances of varying structure, and analyze its behavior, limitations, and tradeoffs relative to problem type.

Progress and evaluation of the project will be assessed in three stages: a project proposal (10%), a project presentation (30%), and a project report (60%).

- Project Proposal - Due Week 6: 1-2 page project description and discussion with the course instructor to ensure sufficient and attainable scope.
- Project Presentation - In class week 10: 15 minute presentation on project + Q&A from classmates and instructor.
- Project Report - Due Week 10: 6-8 page report containing analysis, computation, and discussion of project.

AB, BE, MEng, and MEM students can complete the project in teams of two. PhD students should complete individual projects.

Project description (10%)

he project proposal is a 1–2 page document, prepared in LaTeX, that defines the proposed project and establishes that it has a meaningful and attainable scope. The proposal is due in Week 6 and will be discussed with the course instructor before the project proceeds.

The proposal should address the following elements.

1. **Decision-making problem and motivation.** Describe the decision-making problem and explain why it is worth studying. Identify the application or setting, the decisions that must be made, and the consequences of those decisions. The motivation should answer the question: Why should anyone care about this particular problem or its solution?
2. **Objectives and constraints.** Explain how the decision-making problem will be represented as an optimization problem. Identify the decision variables, objective or objectives, and principal constraints. State the main assumptions needed to make the problem mathematically and computationally tractable.
3. **Proposed solution approach.** Describe how you plan to solve or study the optimization problem. This may include implementing an existing algorithm, using an optimization modeling package or solver, developing a new computational method, reproducing a method from the literature, or comparing multiple formulations or solution approaches.
4. **Expected output and analysis.** Explain what the solution will provide and how its quality or usefulness will be evaluated. Identify the quantities, comparisons, computational experiments, or theoretical properties that you plan to analyze. Be clear about what conclusions the proposed work could support.
5. **Relationship to prior work.** Cite and briefly discuss at least two relevant references. Explain how these sources formulate or solve related problems and how the proposed project builds on, modifies, tests, or departs from that work.
6. **Project contribution.** State what you expect to learn or demonstrate that is not already established by the cited work. The contribution need not constitute a new research result, but the project should do more than reproduce an existing formulation or solver without analysis. For example, the project might study a new application, modify an assumption, compare competing formulations, test sensitivity to problem structure, or identify limitations of an existing method.

A successful proposal should make clear what problem will be solved, how it will be solved, what will be analyzed, and what the project will contribute. The proposed scope should be specific enough to guide the remaining work while remaining feasible within the course timeline.

Project Presentation (30%)

Project presentations will take place during Week 10. Students will sign up online for a presentation time. Each project will be assigned a 20-minute time slot, consisting of a 15-minute presentation followed by 5 minutes for questions and discussion. The presentation should explain the decision-making problem, why it is significant, and any background needed to understand the

project. It should then describe the optimization formulation, solution approach, and main findings. The presentation should make clear what was done, what was learned, and what conclusions are supported by the analysis.

Presenters should plan to use the full 15 minutes. Presentations that are significantly shorter than the allotted time will be penalized. Participation as an audience member is also part of the presentation grade. Students are expected to engage with their classmates' projects and ask thoughtful questions during the discussion periods.

Project Report (60%)

The final project report should be 6–8 pages and follow the IEEE conference paper format. The report must be self-contained: a student who takes this course next year should be able to read the report, understand the problem and solution approach, and appreciate the significance of the findings without having attended the presentation.

The report should use an organization similar to the following:

1. **Abstract.** Briefly summarize the decision-making problem, optimization approach, and principal findings.
2. **Introduction.** Motivate the problem, explain its significance, review the most relevant related work, and state the specific questions or contributions addressed by the project.
3. **Problem Setting.** Define the problem precisely, including the relevant notation, decision variables, objective function, constraints, data, and modeling assumptions.
4. **Solution Approach.** Describe the analytical or computational method used to solve the problem. Include enough detail for the reader to understand how the reported results were obtained.
5. **Results.** Present the main analytical and computational findings. Describe the experimental setup, problem instances, and evaluation criteria needed to interpret the results.
6. **Discussion.** Interpret the results and explain what they reveal about the decision-making problem or solution method. Discuss important modeling choices, limitations, tradeoffs, and unexpected behavior.
7. **Conclusion.** Summarize the main findings and identify reasonable directions for extending or improving the project.

Alternative section titles or organization are acceptable when better suited to the project. Regardless of structure, the report should clearly connect the problem formulation, solution method, evidence, and conclusions.